Shereen Anand

Researcher at Microsoft Research

Email Google Scholar LinkedIn Website

Research Interests

AI4Science, Optimization, Decision-Making, Uncertainity, Game Theory, Computational Neuroscience, Autonomous Agents, LLM Evaluation, Natural Language Processing, Reinforcement Learning, and Multimodal Learning.

Education

Bachelor of Technology in Computer Science and Engineering **Sept 2022 – Jul 2026**
PES University, Bangalore *CGPA: 3.59/5*

KSEAB (12th Grade) **June 2020 – June 2022**
RV PU College *Score: 95.3%*

Experience

Researcher **Oct 2026 – Present**
Microsoft Research (Manager: Gopal Srinivasa) *Bangalore, India*

- Built a **state-coverage metric tool** for C#/.NET using **Roslyn**: parses a repository, builds a call graph (including interface and virtual dispatch), identifies all externally callable entry-point functions, and extracts their parameters, fields, static state, return values and external side effects.
- Designed an **agentic generator–checker–reviewer loop** across two LLM families that abstracts each variable's domain into bounded value classes with executable C# predicates and derives per-function path diagrams and class-level state machines; redesigned the prompt around explicit breadth and depth budgets, shrinking the generated models about 4×.
- Implemented runtime instrumentation through Roslyn source rewriting to measure value-class, path, state and transition coverage during `dotnet test`; showed that fully passing suites reached only 35–40% of input classes, versus 83–90% for extended suites, and surfaced real defects such as missing rollback and uncaught exceptions.
- Evaluated test adequacy on SWE-bench Verified, EvalPlus and kGym, showing that passing the provided tests overstates correctness; built LLM-generated mocks whose state coverage rose from 41% to 99% after feeding missed states back. This work is **under review at ISSTA Main Track 2027**.

Applied Scientist Intern, Central Machine Learning Team **Jan 2026 – Jun 2026**
Amazon (Manager: Kaveri Sivaramakrishnan) *Bangalore, India*

- **Developed ASPEX**, an aspect-grounded persona generation framework integrating hierarchical taxonomy induction with soft assignment, and prompt optimization OPRO technique to model multi-faceted user preferences for personalized conversational recommendation.
- Designed and benchmarked persona-conditioned recommendation across **8 downstream personalization tasks** spanning **Amazon and 3 public datasets** with a cumulative of **300K+ users**, enforcing strict temporal separation between persona construction and downstream evaluation.
- Established state-of-the-art performance over contemporary persona generation baselines, achieving relative **24.2%** PR-AUC and **15.8%** ROC-AUC improvement in the engagement prediction. Observed **18.9%** higher recommendation follow-up rate in suggestions provided.
- Optimized production LLM inference on AWS Bedrock using Claude models, reducing end-to-end latency by 20.3% using an aspect-grounded verification methodology; work submitted to the **EACL Industry Track 2027**.

Student Researcher, Autonomy and Quality for Agents **Jun 2025 – Dec 2025**
Google DeepMind (Manager: Sagar Gubbi) *Bangalore, India*

- **Architected VISTA-Bench**, an adversarial evaluation framework for frontier agentic AI that auto-generates distributionally diverse testing environments using a dynamic Symmetric Adversarial Protocol.
- Engineered scalable evaluation pipelines across **300+ synthetic mobile environments**; formalized automated telemetry metrics for high-dimensional compositional generalization, behavioral trajectory consistency, and multi-step error recovery dynamics, minimizing empirical evaluation overhead by **35%**.
- Identified core brittleness modes in state-of-the-art frontier agents, systematically characterizing performance degradation profiles to prove that despite achieving **90%+ standard benchmark accuracy**, agent policy success rates decayed by up to **42%** under non-stationary covariate distribution shifts.
- Discovered and quantified a multilingual robustness paradox, establishing a statistically rigorous counter-intuitive phenomenon where frontier agents manifest a **22% increase** in out-of-distribution generalization vectors when evaluated using low-resource languages compared to high-resource baselines.

Research Experience

Research Assistant — **Harvard University**
Marks Lab, Advisor: Prof. Debora Marks — *Cambridge, Massachusetts (Remote)*

- Benchmarked **general-purpose language models** against **specialist protein-variant predictors** on shared candidate sets, comparing predicted scores with measured mutation effects.
- Tested whether predictor confidence stayed reliable under harder splits, including **unseen variants within a protein**, held-out proteins and families, and different experimental assays.
- Evaluated calibration on **public mutation-effect datasets**, checking whether low-confidence predictions were less accurate and whether rejecting them improved accuracy on the predictions retained.
- Simulated experiment selection with hidden labels to test whether an uncertainty-based rule picks more informative variants than random selection under a **fixed budget of 20 measurements**.

Research Collaborator — **University of Calgary**
DENOS Lab, Advisor: Prof. Steve Drew — *Calgary, Canada (Remote)*

- Extending **EDSim**, an LLM-agent emergency department simulator where GPT-4o-mini-driven patient, triage nurse, bedside nurse and physician agents run the full care journey.
- Evaluating simulated wait times against historical data from a Level-1 trauma centre, stratified across **5 CTAS acuity levels**, with baseline outputs matching historical wait-time distributions.
- Running what-if surge experiments over 326 simulated patient visits (187 baseline, 139 surge) with a 60% arrival-rate increase, quantifying a **75.4% rise** in mean ED length of stay (552 to 968 minutes).
- Analyzing stage-level bottlenecks across a 79-bed, 15-physician ED replica, showing arrival-to-assessment times up **74.7%** and post-disposition boarding up **100.7%** under stress.

Research Intern — **IIIT Bangalore**
Advisor: Prof. V. Ramasubramanian, Summer Research Internship Program — *Bangalore, India*

- Constructed low-resource multilingual ASR systems leveraging **matching networks** and **self-supervised learning** to address data scarcity in underserved linguistic communities.
- Engineered cross-lingual transfer mechanisms using **wav2vec 2.0** with alignment-aware modeling, reducing **phoneme error rates by 28%** in complex code-switching scenarios.
- Analyzed phonetic confusion patterns across **15 language pairs**, identifying systematic misalignments in acoustic feature representations that propagate under extreme data scarcity (**<100 hours** labeled audio).
- Demonstrated that **self-supervised pretraining** can democratize speech technology access for **11 low-resource languages**, establishing baselines for future language-specific acoustic modeling.

Open-Source Contributions

Nanda Town (MIT Media Lab). Developing a proof of concept, comprising a protocol, a plugin and a test, for *Nanda Town*, an open-source sandbox for building and testing protocols for AI agents across twelve layers, including identity, trust, negotiation and payments. Working with the NANDA team led by **Prof. Ramesh Raskar**.

Digital Twin for AR Microsurgery (Blevins Lab, Stanford University, Hassanpour Lab, Dartmouth College). Designed a phantom and simulation study of when an AR surgical overlay supports accurate spatial decisions and when it misleads, comparing no overlay, a fixed-confidence overlay and an uncertainty-aware overlay under varied registration error, occlusion and clutter, and measuring localisation error, unsafe proximity decisions and confidence calibration. Guided by **Dr. Nikolas Blevins (Stanford University)** and **Prof. Saeed Hassanpour (Dartmouth College)**.

Selected Publications

Anand, S., Wang, J., Du, Y. What Structurally Breaks an Agent? Auditing Difficulty in Stateful Tool-Use Workflows. [arXiv]

Anand, S., Singh, C. Verification Allocation in Stylized LLM Agent Pipelines under Hidden Tool Degradation: A Discrete-Time Mean-Field Analysis.[arXiv]

Padmanabhan, V., **Anand, S.**, Peddi, V., Vamshi, V. Enhanced Trading Systems with Intelligent Market Analysis. In: *Proceedings of the 4th International Conference on Artificial Intelligence and Applications (ICAIA)*. **Accepted**. [Paper]

Anand, S. Identifying Learning Rules from Representational Plasticity. [Preprint]

Mourya, A., **Anand, S.**, Mukerjee, R., Puranik, V., Kaveri, S. ASPEX: Aspect-Grounded Persona Generation with Hierarchical Taxonomy (Amazon). In: *EACL Industry Track 2027*. **Under review**. [Paper]

Anand, S. Tumour accumulation is not tumour-to-liver selectivity: a publication-aware polymer nanoparticle reanalysis. [ChemRxiv]

Anand, S., Srinivasa, G., Mohan, S., Rastogi, A., Prasad, A. GHOST: Hidden-Test Coverage Evaluation of Coding Agents on SWE-bench Verified (Microsoft Research). In: *ISSTA 2027*. **Under review**.

Im, C., **Anand, S.**, Zhou, Y. Conditional Generative Learning in Metric Spaces (equal contribution). In: *ICML 2027 (Main Track).* **Under review**.

Anand, S. Estimation Robustness of Static Congestion Pricing Under Learned Demand. [Preprint]

Wu, J., Ledingham, H., Wang, Z., **Anand, S.**, Teitge, B., Burn, A., Ouadihi, O., Ghattas, M., Vicaldo, D., Cociuba, S., Harmon, M., Chowdhury, T., Marshall, Z., Williamson, T., Risling, T., Lang, E., Zhou, J., Holodinsky, J., Drew, S. EDSim: An Agentic Simulator for Emergency Department Operations. In: *ICML 2027 (Main Track).* **Under review**.

Anand, S. Publication-block calibration reveals wide uncertainty in PLGA release endpoint prediction. [ChemRxiv]

Projects

Mean-Field Game Model of LLM Agent Teams | *Python, AutoGen, Optimisation* **2026**

- Proposed a **mean-field game** model of multi-agent LLM teams, treating each agent as facing the population share of agents in each role, to replace the many-agent problem with one representative-agent problem.
- Designed a validation plan comparing the real **AutoGen** team against the mean-field prediction for team sizes **N = 2, 4, 8, 16**, with the prediction error measured by total variation distance over matched runs.
- Planned an **equilibrium** test in which one agent deviates to its best alternative strategy, reporting the score gain as a lower bound on how exploitable the team is.

Robustness of Brain-Aligned Vision Models | *Python, PyTorch, Brain-Score* **2025**

- Proposed a **brain-alignment stress test** to investigate whether vision models that better predict primate visual-cortex responses also retain more human-like recognition under controlled **blur, occlusion, and contrast changes**.
- Designed a comparison of **feedforward and recurrent vision models**, measuring accuracy, confidence calibration, and image-level error patterns across degradation strengths to separate overall robustness from similarity to biological vision.
- Planned a joint evaluation of **neural predictivity and behavioral alignment**, using matched stimuli where suitable reference data are available, to test whether recurrence improves agreement with human errors beyond gains in recognition accuracy.

Deforestation Detection & Forecasting from Satellite Imagery | *Python, TensorFlow, GDAL, U-Net* **2024**

- Aggregated and preprocessed multi-temporal Sentinel-2 and Landsat imagery across **12 districts**, applying cloud-masking and NDVI-based filtering to construct a clean, geo-referenced training corpus.
- Trained a **U-Net segmentation model** to delineate forest cover loss at pixel level, reaching **89% IoU** on held-out regions through iterative tuning of encoder depth and loss weighting for class imbalance.
- Formulated an LSTM-based forecasting layer atop segmentation outputs to project deforestation trends 6 months ahead, cutting mean prediction error by **21%** against a linear-regression trend baseline evaluated on the same held-out districts.

Skin Cancer Detection using Deep Learning | *Python, PyTorch, CNN* **2023**

- Curated and preprocessed a dermoscopic image dataset of **10K+ lesion samples**, applying targeted augmentation (rotation, color-jitter, oversampling) to correct class imbalance in minority melanoma cases.
- Fine-tuned a **CNN classifier** (EfficientNet backbone) via transfer learning to distinguish malignant from benign lesions, attaining **92% classification accuracy** and **0.90 F1-score** on a stratified test split.
- Applied **Grad-CAM-based interpretability** to visualize activation regions driving predictions, cross-validating model attention against dermatologist-annotated lesion boundaries to check for spurious correlation.

Teaching & Mentorship

Student Mentor **IEEE RAS**
PES University *2023–2024*

- Guided junior students through robotics competition projects and provided technical mentorship. Conducted workshops on robot programming.

Teaching Assistant **Practical Applied Machine Learning**
Centre for Entrepreneurship and Innovation, PES University *2023*

- Designed lab exercises and practical sessions for 56 students covering ML concepts. Led hands-on tutorials implementing ML algorithms provided one-on-one guidance on project development and debugging.

Founder, Metatron **Government Schools of Karnataka, Bangalore**
Machine learning workshops for girls *2021–2022*

- Founded and ran Metatron, an after-school machine learning program for girls in local government schools across Bangalore, reaching 250 students across 6 schools over 100 hours of instruction.

Technical Skills

Programming Languages: Python, C++, R, MATLAB, SQL, Java, JavaScript, TypeScript, HTML/CSS
Machine Learning & AI: TensorFlow, PyTorch, scikit-learn, Keras, XGBoost, LightGBM, reinforcement learning, neural networks, ensemble methods, feature selection, hyperparameter optimization
Data Engineering: Apache Spark, Hadoop, Airflow, Pandas, NumPy, SciPy, real-time streaming, event processing, data quality checks, feature stores
Visualization & Analytics: Matplotlib, Seaborn, Plotly, Tableau, Power BI, interactive dashboards
Backend Development: Python, C++, Java, microservices, REST APIs, GraphQL, message queues
Cloud & Infrastructure: AWS (EC2, S3, Lambda, RDS, SQS), GCP (BigQuery, Dataflow), Azure, Docker, Kubernetes, Terraform, real-time data pipelines
Databases: PostgreSQL, MySQL, MongoDB, Redis, Cassandra, InfluxDB, KDB+/q, time-series databases
System Design: High-frequency trading infrastructure, low-latency systems, distributed computing, microservices, load balancing, caching, fault tolerance
Development Tools: Git, GitHub, VSCode, IntelliJ, Jupyter, Docker, CI/CD, testing frameworks (Pytest, Jest), profiling
Operating Systems: Linux, Windows, macOS, Shell scripting, system administration

Achievements & Awards

1. Runner-Up – Amazon Internal Hackathon 2026
2. Top-100 – Google Girl Hackathon 2025 Software Round
3. Millennium Fellow for the Class of 2023 – United Nations Academic Impact and Millennium Campus Network
4. Finalist – Smart India Hackathon in Software Edition 2023, selected at national level.
5. Runner-Up – Robowars Tournament in 2023
6. Finalist – Srinivasa Ramanujan Mathematics Competition (SRMC) 2016, invited to offline finals at IIT Bombay.